Hurdle Bayesian model: main changes from the article

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Same structure as the article: a probit BART for the existence of a flow, and a (zero-truncated) negative binomial with log-mean η_{ijt} for its volume. Y_{ijt} is the flow $i \to j$ in period t. The two main changes are in the volume: migrant stocks (Section 1) and country-period shocks (Section 2).

1 Two new covariates: the migrant stocks

$S_{i\to j,t}$ is the number of people born in i who live in j at mid-t (UN DESA 2020,). Two stocks enter the covariates x_{ijt} of the volume, and BART too:

- **Diaspora stock** $\log(1 + S_{i\to j,t})$: people from the origin who already live in the destination. It is the network effect: new migrants go where their family and friends already are, and get help for a home, jobs and information. Coefficient +0.14 [0.12, 0.17].
- **Reverse stock** $\log(1 + S_{j\to i,t})$: people born in the destination j who live in the origin i, so it is the potential returns. A person born in Algeria who lives in France counts in the reverse stock of the corridor France → Algeria. We expect: a high reverse stock potentially leads to high returns. Coefficient +3.0 [2.94, 3.05], by far the largest.

2 Country-period shocks

The volume mean now Thestructural level of a corridor is

$$\mu_{ijt} = \alpha_{it} + \gamma_{jt} + \beta^\top x_{ijt}, \qquad \alpha_{it} = c + \tau_\alpha a_i + \theta_\alpha z_{it}, \qquad \gamma_{jt} = \tau_\gamma g_j + \theta_\gamma z_{jt}. \tag{1}$$

The country effects keep a structural part a_i, g_j, fixed in time, and an economic part θz_{it}, with z_{it} the log GDP of year $t-1$, standardised across countries. In the article they were fixed in time, with the GDP of 2010. For a corridor open in $t-5$, the log-mean of the window is equation 5 of the article plus the new shocks:

$$\eta_{ijt} = \underbrace{(1-\rho_{ij})\,\mu_{ijt}}_{\text{long-run level}} + \underbrace{\rho_{ij}\,\omega\,\log Y_{ij,t-5} + \rho_{ij}\,(1-\omega)\log Y_{ij,t-10}}_{\text{past flows: AR(2)}} + \underbrace{\xi_{it} + \zeta_{jt}}_{\text{shocks}}. \tag{2}$$

(unfolded equation 5 + new shocks). A corridor closed in $t-5$ has $\eta_{ijt} = \mu_{ijt} + \kappa_k + \xi_{it} + \zeta_{jt}$, with κ_k trained on emergent corridors, M49 sub-region k (e.g, West-Europe, North America,...). α_{it} and γ_{jt} are inside μ: they give the long-run level, and in one window they act only with weight $1-\rho \approx 0.08$.

Why In a crisis (war, economic collapse, new policy) all the flows of a country move together. Without shocks the corridors are independent given the parameters. We first tried shocks inside α and γ, but these runs did not converge. We prefer the following version: the shocks ξ_{it} and ζ_{jt} directly in η. They act fully in their window period, and after that only through the lagged flows.

How ξ and ζ are estimated One shock ξ_{it} per origin and period (and one ζ_{jt} per destination and period), but only two variances for all of them: $\xi_{it} \sim \mathcal{N}(0, \sigma_\xi^2)$, $\zeta_{jt} \sim \mathcal{N}(0, \sigma_\zeta^2)$. Stan samples everything together by HMC: the shocks, their two variances. Each shock is learned from the flows of its country in that period, and σ_ξ, σ_ζ learn from the dispersion between all the shocks. They are well identified: $\sigma_\xi = 0.65$ [0.62, 0.69] and $\sigma_\zeta = 0.45$ [0.42, 0.47], far from zero. For 2015, new shocks are drawn in each posterior draw, shared by all the corridors of a country.

3 Results for 2015

					Coverage (95%)	
	FP	FN	MAE	MAPE	all	open corridors
September Article	285	4,797	1,135	54.2%	96.1%	93.8%
Now	26	3,820	1,121	29.5%	97.4%	95.0%

Article: 190 countries (35,910 corridors). Now: 187 countries (34,782 corridors): we dropped Guadeloupe, French Guiana, Martinique and Réunion (no GDP in WDI, not in V-Dem) and added back Timor-Leste (we found all of its data).

Reminder: the volume model

Likelihood (positive flows only, zero-truncated NB2):

$$P(Y_{ijt} = y) = \frac{\mathrm{NB}(y \mid m_{ijt}, \phi_{ijt})}{1 - \left(\frac{\phi_{ijt}}{\phi_{ijt}+m_{ijt}}\right)^{\phi_{ijt}}}, \quad y \geq 1,$$

$$m_{ijt} = e^{\eta_{ijt}}, \qquad \mathbb{E}[Y_{ijt}] = \frac{e^{\eta_{ijt}}}{1 - \left(\frac{\phi_{ijt}}{\phi_{ijt}+e^{\eta_{ijt}}}\right)^{\phi_{ijt}}}.$$

Mean: equation 5 plus the country-period shocks

$$\mu_{ijt} = \alpha_{it} + \gamma_{jt} + x_{ijt}^{\top}\beta, \qquad \eta_{ijt} = \begin{cases} (1-\rho_{ij})\,\mu_{ijt} + \rho_{ij}\,\bar{L}_{ijt} + \xi_{it} + \zeta_{jt}, & Y_{ij,t-5} > 0, \\ \mu_{ijt} + \kappa_k + \xi_{it} + \zeta_{jt}, & Y_{ij,t-5} = 0, \end{cases}$$

$$\bar{L}_{ijt} = \omega \log Y_{ij,t-5} + (1-\omega)\log Y_{ij,t-10} \qquad (\omega = 1 \text{ if } Y_{ij,t-10} \text{ is 0 or missing}).$$

Coefficients from last simulation Covariates are standardised except the binary ones (border, language, colonial link). These are long-run effects: in one period a covariate moves η only by $(1-\rho_{ij})\beta$.

	Median	95% interval
log distance	−0.151	[−0.176, −0.127]
land border	−0.174	[−0.283, −0.067]
official language	+0.237	[0.191, 0.285]
colonial link	+0.052	[−0.017, 0.113]
democracy of the origin $(t-5)$	−0.170	[−0.271, −0.083]
conflict intensity of the origin $(t-5)$	+0.074	[0.039, 0.115]
diaspora stock	+0.144	[0.120, 0.170]
reverse stock	+2.999	[2.939, 3.050]